Titanic EDA Analysis

Dataset Information

- Total passengers: 891
- Total columns: 12

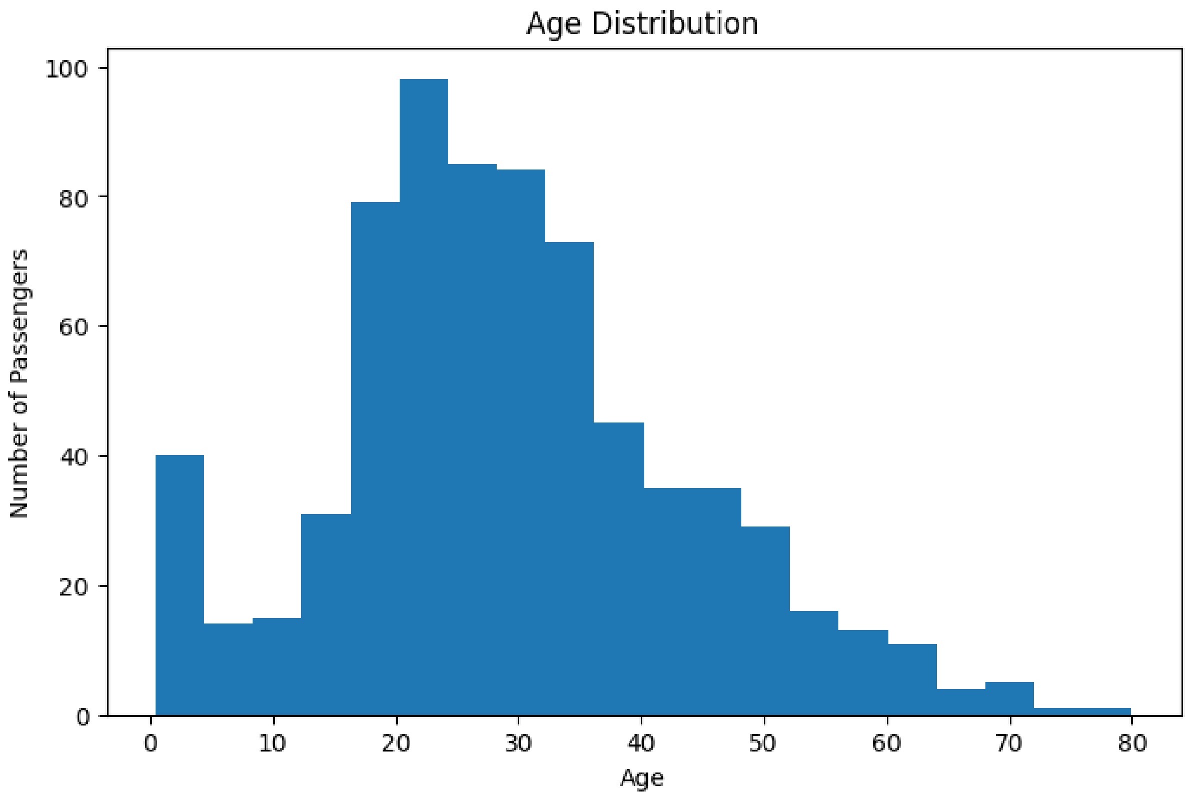
Findings

- Missing values found in Age, Cabin, and Embarked.
- Most passengers were between 20 and 40 years old.
- Some passengers paid very high fares.
- Female passengers survived more often than males.
- Passenger class affected survival.

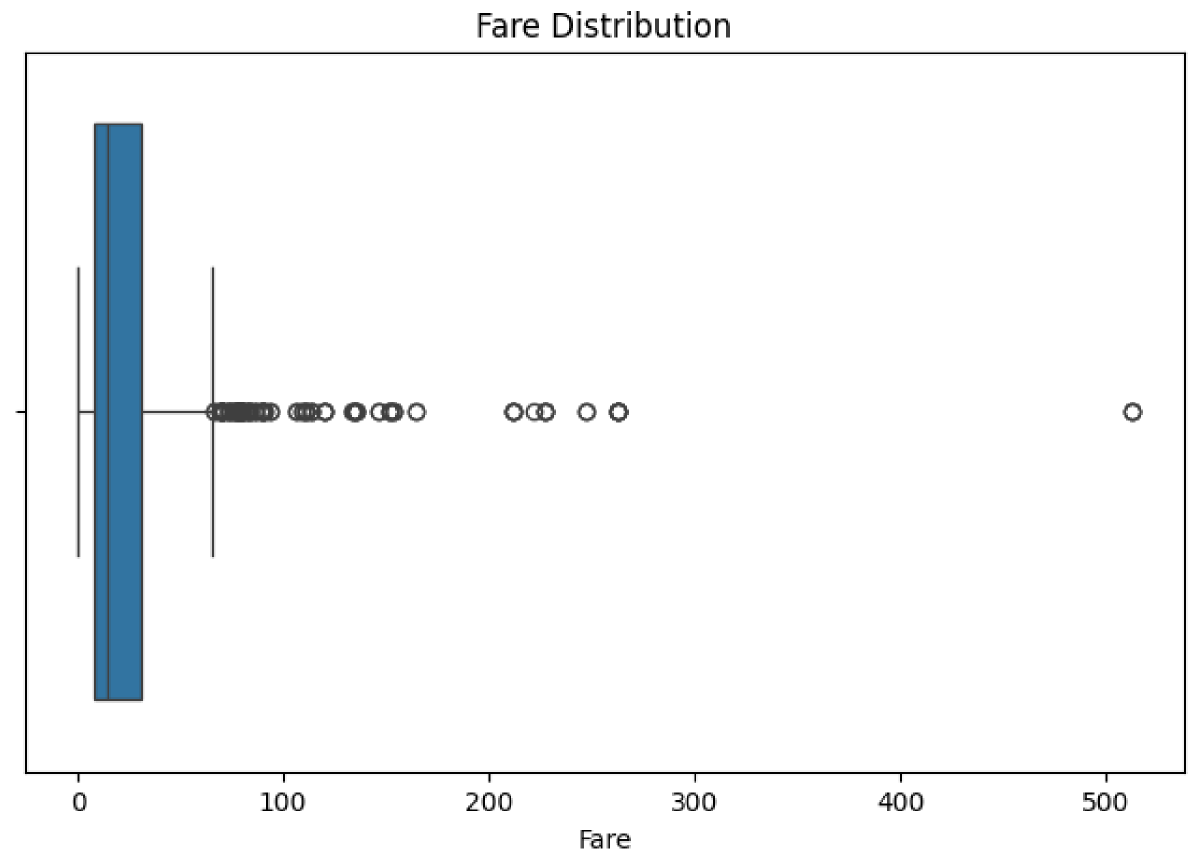
Conclusion

The Titanic dataset was analyzed using graphs and statistics. Age, gender, fare, and passenger class influenced survival.

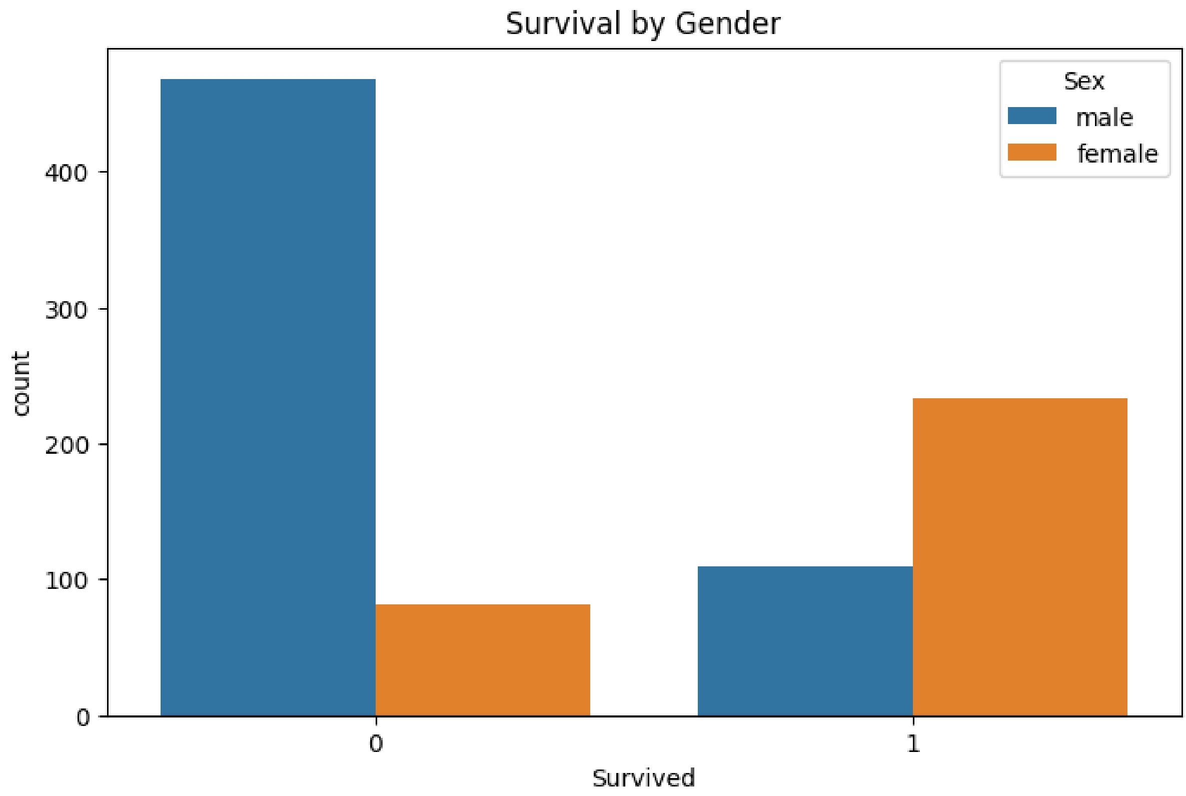
```
import matplotlib.pyplot as plt
plt.figure(figsize=(8,5))
plt.hist(df['Age'], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```



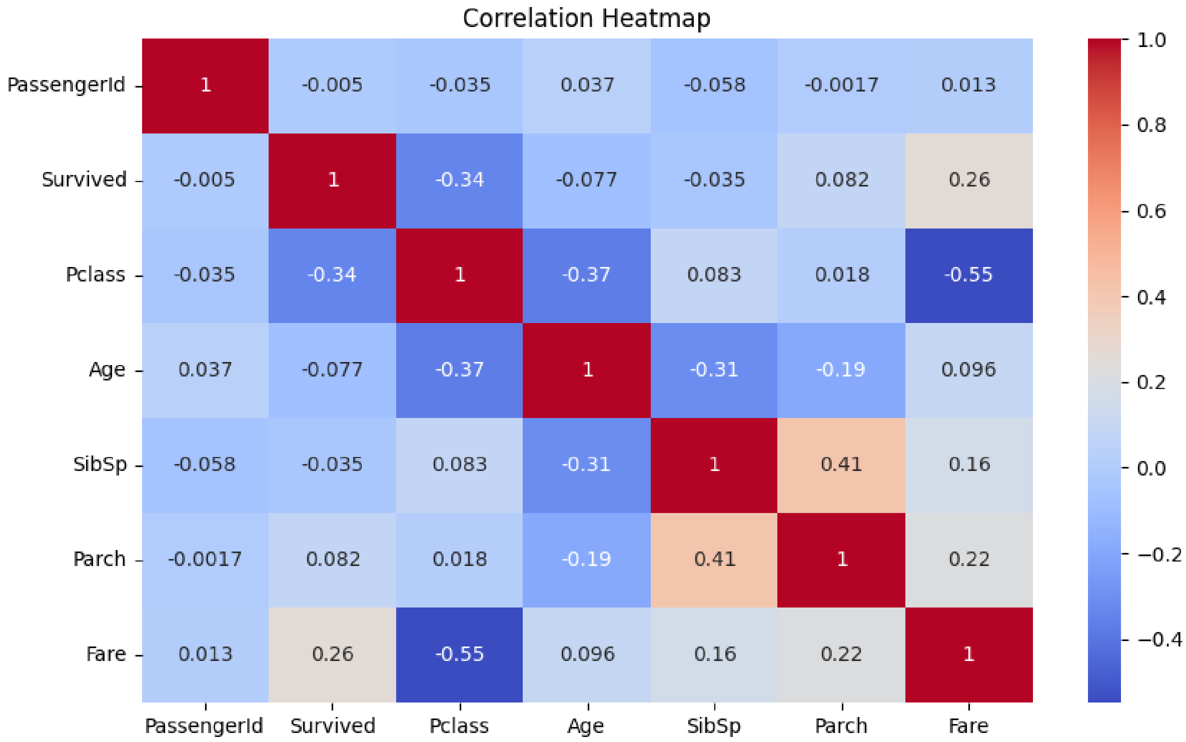

```
import seaborn as sns
import matplotlib.pyplot as plt
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Fare'])
plt.title("Fare Distribution")
plt.show()
```



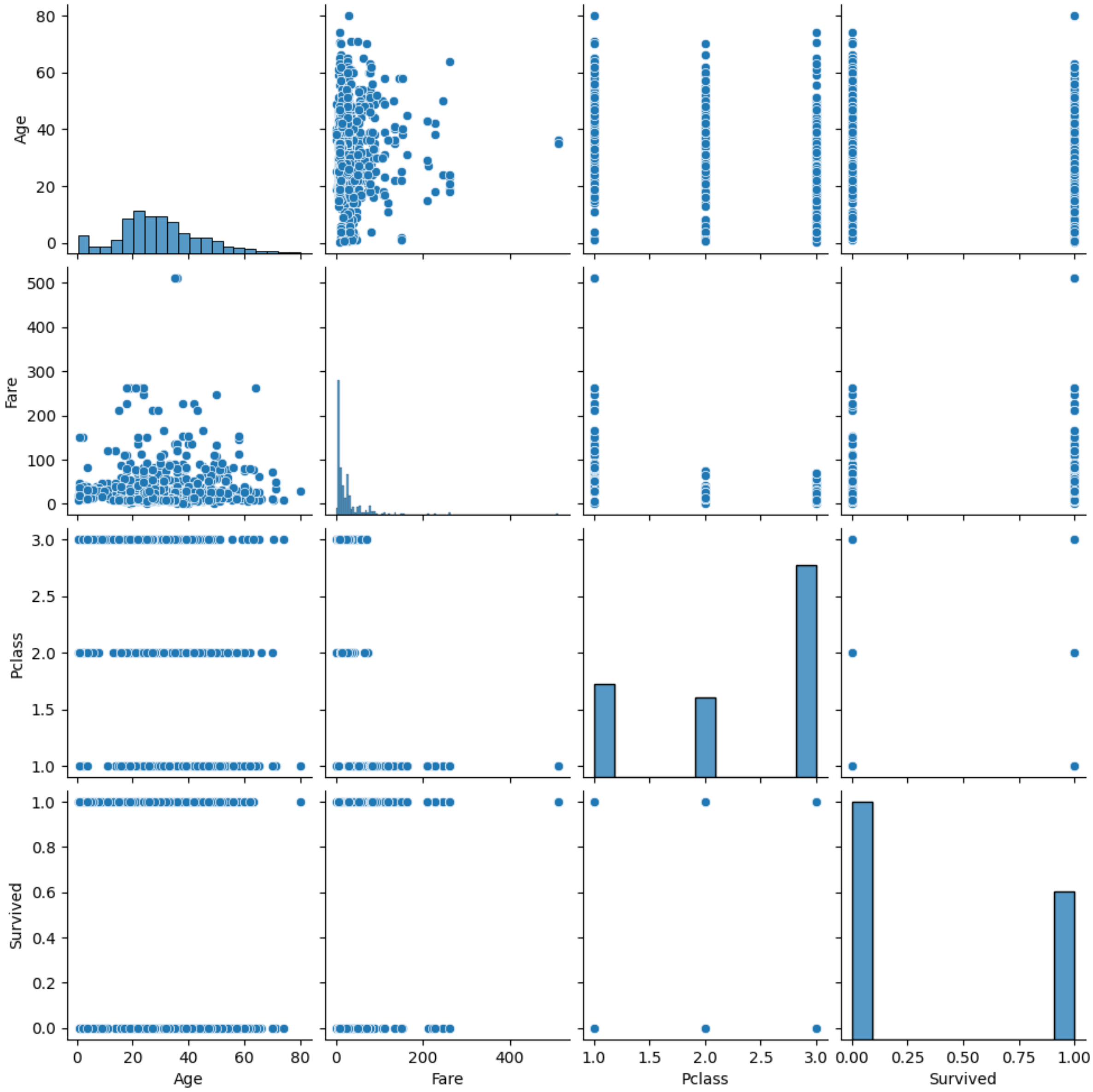

```
plt.figure(figsize=(8,5))

sns.countplot(x='Survived', hue='Sex', data=df)

plt.title("Survival by Gender")

plt.show()
```



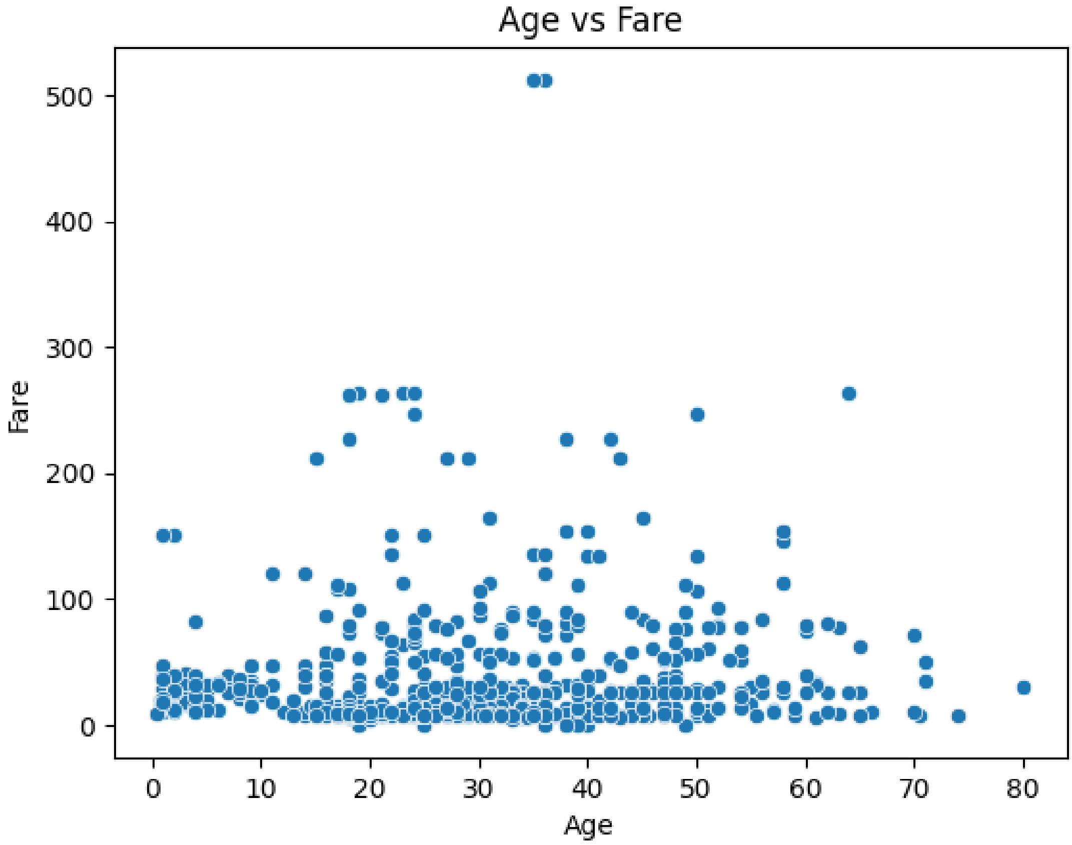

```
plt.figure(figsize=(10,6))
sns.heatmap(
   df.corr(numeric_only=True),
   annot=True,
   cmap='coolwarm'
)
plt.title("Correlation Heatmap")
plt.show()
```




```
sns.pairplot(
  df[['Age','Fare','Pclass','Survived']]
)
plt.show()
```




```
import seaborn as sns

import matplotlib.pyplot as plt

sns.scatterplot(x='Age', y='Fare', data=df)

plt.title("Age vs Fare")

plt.show()
```


Age vs Fare
500
400
300
200
100
0
Fare
0
10
20
30
40
50
60
70
80
Age